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Unit 4 - Dive into Flow

PDF Documents

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UNIT 4 DIVE INTO FLOW

● The Magical Playground

Imagine a playground where everything seems like magic. You step inside, and suddenly the swings move on their own, the slides light up, and the see-saws know when to lift and lower. But here's the secret—it's not magic at all! It's technology working behind the scenes to create a playground like you've never seen before.

A PLAYGROUND THAT THINKS!



In this magical playground, there are special sensors that help the equipment "think." These sensors can detect if someone is near or if it's safe to use certain parts of the playground. For example:

THE TALKING SWING As you sit on the swing, it starts to move gently, but it can also "talk" to you! It might say, "Hold on tight!" or "Swing higher!" depending on how fast you're going.

**THE FRIENDLY SEE-SAW**

This see-saw knows when you're ready to play. It moves up and down smoothly, making sure that both sides are balanced so that everyone has fun!

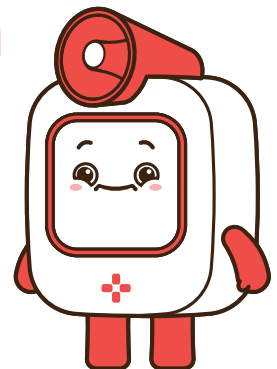
THE PLAYGROUND THAT LEARNS

What's even cooler is that this playground can learn from how kids play. It remembers which parts of the playground are the most popular and adjusts itself to make sure everyone gets a turn. If there's a big line for the slide, the playground might suggest games to play while you wait!

FUN AND SAFE FOR EVERYONE

The magical playground isn't just about fun; it's also about keeping everyone safe. The equipment has sensors that make sure no one gets hurt. The slides won't let you go if there's someone at the bottom, and the swings slow down if you're going too fast. It's like the playground is looking out for you!

Let's imagine your own playground!



Multiple Choice Questions

1 What is a Sensor?

Circle the correct answer: A sensor is a device that:

- ☐ A Tells time
- ☐ B Detects things around it
- ☐ C Plays music

CORRECT ANSWER

☒ A, ☒ B

2 What is a countdown display?

- ☐ A A device that shows how fast you can run
- ☐ B A device that increases time for an event
- ☐ C A device that shows numbers decreasing to zero
- ☐ D A device that only shows letters

CORRECT ANSWER

☒ C

3 Which module can sense the distance

- ☐ A ToF
- ☐ B Button
- ☐ C IMU
- ☐ D Speaker

CORRECT ANSWER

☒ A



●● Reflection Questions

- 1 Imagine you are building a Smart Slide. What will the distance sensor do if someone is standing too close to the bottom of the slide?

- 2 Can you think of another place where sensors help keep people safe? Draw or write about it below:

- 3 What are the primary reasons why humans developed systems for measuring time? Also, what are some of the advantages of accurate timekeeping in modern society?